



EXAMENSARBETE INOM TECHNOLOGY,
GRUNDNIVÅ, 15 HP
STOCKHOLM, SVERIGE 2017

An Analysis of Climate Change and its Effects on the Electricity Generation

Supporting a CLEWs assessment in Ghana

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TRITA -IM-KAND 2017:22

Abstract

Sustainable, effective and affordable access to energy plays a significant role when it comes to improving people's living conditions, and supporting human and economic development. Therefore, the access and availability of electricity is an important factor and indicator of how well developed a country is. Furthermore, the generation of electricity is often strongly interlinked with other sectors such as climate, land use, and water. All these resources' systems can have a certain effect on the energy production and vice versa. It is therefore a complex sector, and it needs to be developed with these interlinkages in mind.

The aim of this project is to study and analyze the future electricity system in Ghana and its linkages to climate change and water resources. In this project, the modelling tools focus to optimize the electricity generation in consideration of economic values, as well as analyzing climate change and its impact on electricity generation by hydropower.

To make a projection of the electricity system in Ghana, two modelling tools were used in the study, OSeMOSYS and ONSSET. The results generated by the models give an indication of potential optimal generation mixes between 2015 and 2050, for the different set of considerations on which the scenarios were built. The projected electricity generation consists of a combination of both renewable energy technologies and fossil fuel dependent technologies. The contribution from concentrated solar power (CSP) stands for the largest share of the mix in the long term. The results also suggest that climate change most certainly will have an impact on the electricity production in Ghana, due to its effect on the performance of hydropower production. This is an indication that Ghana's reliance on hydropower for its electricity production, could not be sustainable and reliable in the long term.

Ghana is an unequally developed country and there are still many undeveloped regions with poor access to electricity, especially in the northern parts of the country. There is also a rising demand of water and land for agricultural use. Since these resources also are important for the sustainable production of electricity, further and more detailed investigation of the conflicts between these sectors is necessary for a sound and compatible sectoral development and sustainable management of resources.

Sammanfattning

Hållbar, effektiv och prisvärd tillgång till energi och elektricitet är en viktig del i att förbättra människors livsvillkor och för stödja mänsklig och ekonomisk utveckling. Tillgången till elektricitet är på grund av detta en viktig indikator på hur väl utvecklat ett land är. Vidare är produktionen av elektricitet stark sammankopplad med andra sektorer såsom klimatet, landanvändning och vatten, då alla dessa resurser påverkar energiproduktionen och vice versa. Energisektorn är därför mycket komplex och utvecklingen av den kräver att dessa kopplingar till andra sektorer respekteras.

Syftet med det här projektet är att studera och analysera det framtida elsystemet i Ghana samt dess kopplingar till klimat och vatten. Projektets fokus är att optimera elproduktionen utifrån kostnad och att analysera klimatförändringarnas påverkan på elproduktionen från vattenkraft.

För att kunna skapa en modell över Ghanas elsystem i framtiden användes två modelleringsverktyg i projektet, OSeMOSYS och ONSSET. Dessa genererade ett resultat av den optimala fördelningen av produktionstekniker inom elproduktionen under åren 2015 till 2050. Resultatet visar att den mest kostnadsoptimala elproduktion består utav en blandning av förnyelsebar och fossilberoende el. Den största delen av produktionen under den senaste delen av modellen kommer från koncentrerad solenergi. Vidare framgår det av resultatet att klimatförändringar kan komma att påverka elproduktionen i Ghana då vattenkraftens genereringsförmåga är starkt klimatberoende och försämras i torrare klimat. Detta är en indikator på att Ghanas beroende av vattenkraft för sin elförsörjning inte är långsiktigt hållbart.

Ghana är ett land med en ojäm utvecklingsgrad och det finns fortfarande många outvecklade områden med bristande tillgång till elektricitet. Dessutom ökar ofta behovet av vatten och mark för jordbruk i dessa områden och då dessa resurser även är viktiga för elproduktionen är det av största vikt att fortsätta studier undersöker eventuella konflikter mellan dessa sektorer för att undvika negativa synergieffekter i framtiden.

Acknowledgement

We would like to start by expressing our gratitude to our supervisors Eunice Ramos and Vignesh Sridharan, for always supporting and helping us with our study by sharing their experience and expert knowledge with us. Without whom the execution of this project would not have been possible. We would also like to express our appreciation to Mark Howells for enabling us to be part of this project.

We also want to thank the organization of SADA for agreeing to host us and providing such a comfortable office space for us in Ghana. A special thank you to Charles Abugre, Christiana Akpilima-Atibil, Habiba Nantuo, Millie Ashkar and Eric Ewoh for generously hosting us and always helping us with a diversity of practical things.

Finally, we would like to express a warm thank you to Amber Ahmed and Jindan Gong for sharing this wonderful experience in Ghana with us, and supporting us in our struggles with the project.

Acronyms and abbreviations

BAU	Business as Usual
CF	Capacity factor
CLEWs	Climate, Land, Energy and Water strategies
CO _{2,eq}	Carbon dioxide equivalent
CSP	Concentrated Solar Power
FAO	Food and Agriculture Organization of United Nations
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GIS	Geographical Information Systems
GPRS	Ghana Poverty Reduction Strategies
KTH-dESA	KTH - division of Energy Systems Analysis
LCOE	Levelised cost of electricity
MG	Mini-grid
MoManI	Model Management Infrastructure
NSEZ	Northern Savannah Ecological Zone
ONSSET	Open Source Spatial Electrification Toolkit
OSeMOSYS	Open Source Energy Modelling System
PJ	Petajoule
PP	Power plant
PV	Photovoltaic
RE	Renewable energy
RES	Renewable Energy Sources
RefES	Reference Energy System
RET	Renewable Energy Technologies
SA	Stand-alone
SADA	Savannah Accelerated Development Authority
SDGs	Sustainable Development Goals
SNEP	Strategic National Energy Plan
TES	Thermal energy storage
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
UN DESA	United Nations Department of Economic and Social Affairs
WHO	World Health Organization

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1. Introduction

Sustainable, effective and affordable access to energy plays a significant role when it comes to improving people's living conditions and supporting human and economic development. A great challenge for most developing countries is to develop the energy system and increase the electricity access. Part of the challenge is to provide energy access for the total population of a country, while also taking into account the interlinkages between other sectors. Too often there has been a tendency to ignore the interactions between the different sectors and across national borders. Ignoring these interlinkages can have the effect that success in one sector or location causes problems and comes at the expense of a different sector or location. The need of land for energy infrastructure and the energy demand for agriculture, are two examples of how energy is closely interlinked with other sectors. With population growth and increasing life standards, the world's key resources will be under an increased stress. Today, an extensive use of natural resources like freshwater and wood, leading to the depletion of forest cover, and increased greenhouse gas emissions have become an accelerated problem. Since these problems often comes as a result of improved living conditions, the development needs to take place with consideration of the local environment and the effects on the climate. Many developing countries today are facing the struggle of developing their energy system in a sustainable and efficient way to enable economic growth in the country, and to increase the quality of life for the population (UNDESA Division for Sustainable Development, 2014).

There is a clear relationship between poverty and absence of adequate energy services such as heat, mechanical power and light. Therefore, an essential part for developing countries to lift people out of poverty, is the achievement of sustainable energy services (Ghana Energy Commission, 2012). Ghana, is one of these developing countries and this thesis presents the study of an integrated system analysis between electrification and climate change and the environment with Ghana as a case study.

2. Aim and Objectives

The scope and objectives of this study are presented in this section. This section also informs on the main modelling tools used in the project, and closes with the three research questions proposed to be investigated in the study.

2.1 Aim

The aim of this project is to investigate the development of the electricity system of Ghana up to 2050 and the achievement of 100% electrification access for the non-electrified population. The aim will be achieved through the development of models to represent the electricity system of the country. Two modelling tools will be used for this purpose, both of which have been developed at KTH-dESA. The models will allow to investigate how the electricity system could evolve under different scenarios, and what would be the least-cost options to supply electricity to households in Ghana. The electricity sector analysis will be tailored, to the extent possible, to include elements from other systems, such as water and climate. In this way, the study will incorporate aspects from the CLEWs framework as it will be

providing insights over other sectors besides the energy sector. The aim is also to develop the understanding of the different challenges the country has, linked to sustainable development and sustainable use of resources such as energy and water as well as the resources linkages with climate change. To do so, two modelling tools used for long-term energy planning will be used to investigate potential development futures of the electricity sector of Ghana: the Open Source energy Modeling System (OSeMOSYS) and the Open Source Spatial Electrification Toolkit (ONSSET). This work is foreseen to contribute as an initial effort to the development of an integrated resource assessment of Ghana.

2.2 Objectives

The overall objective of this project is to develop a model of the electricity system in Ghana in order to investigate:

- 1) the deployment of renewable energy technologies (RET) in the generation mix;
- 2) estimate the differences in greenhouse gases (GHG) emissions for different climate futures;
- 3) Explore the potential impact of climate change in hydropower generation and the implications to the national electricity generation mix.

To get a full picture of the situation in the country, the project will look at the points where the energy sector and the climate system interact with each other. To analyze how these sectors depend and interrelate, a model of the energy system will be developed and the linkages to the other sectors will be established through different scenarios. The scenarios are designed to explore the renewable energy potential of the country, investigate the impact of climate change in hydropower generation for different climate futures, and perform a comparative analysis of the electricity sector's GHG emissions across scenarios.

2.4 Research questions

One of the main focus of this thesis is the investigation of the deployment of RET in Ghana. Several sources confirm the renewable energy potential of the country. In the master plan of Savannah Accelerated Development Authority (SADA) it is pointed out that there is plenty of renewable energy potential such as solar energy in the Northern Savannah Ecological Zone (NSEZ). There are also ongoing power projects of solar farms and new hydropower plants in NSEZ, to strengthen and increase the electricity supply (SADA p.57, 2016). Thus, the use of renewable energy technologies could be particularly interesting to supply electricity to remote areas or to consumers not connected to the grid at present. The government has inclusively promoted initiatives in this line such as 30% of electricity supply in NSEZ should be produced by renewable energy technologies (SADA p.104, 2016). Thereby it is important to investigate the role of renewable energy sources (RES) potential in the development of the electricity sector of Ghana, and also in the achievement of the SDG7 (more specifically target 7.1: to ensure universal access to reliable, affordable and modern energy services by 2030). Thus, during the development of this project, the study aims at answering the following research questions:

- Could large scale investments in RET such as solar and wind energy prove to be economically feasible for Ghana in the medium-term, to power the electricity system of the country under

different climate futures? And would such investments support the climate resilience of the electricity generation system in Ghana?

- What would be the contribution of such RET investments in the achievement of the sustainable development goal 7.1, related to ensuring universal access to affordable, reliable and modern energy services to all, by 2030?
- Could the expansion of hydropower generation be a solution for the electricity sector development in Ghana? What would be the economic implications and how could hydropower generation be affected under different climate futures?

3. Ghana in context

To have a good knowledge of the situation in the country studied and to explore how different circumstances may affect the evolution of the electricity sector it is vital to perform a background check of the country context in terms of socio-economic development and also at a sectoral level. The following section is dedicated to present the relevant background information about Ghana, necessary to perform the different steps described in the methodology as well as to understand and to analyze the results from this study.

3.1 Socio-economic background and trends

Ghana, officially the Republic of Ghana, formerly the Gold Coast, is a multicultural nation located in West Africa. In 2015, Ghana was measured spanning a land area of 227,540 square km and the same year the country's population reached 27.4 million people, with an annual population growth rate of 2.3 % in 2014 to 2015 (The World Bank, 2016). The country was a British colony until 1957, when it became the first sub-Saharan African nation to declare its independence. Ghana ranks among the top three countries in Africa considering freedom of speech and freedom of the press and the country is today considered to be a stable democratic state and a model for Africa as a whole (The World Bank, 2017). The GDP of Ghana has slowed down to 3.7% in 2015 because of several economic challenges, including the low price for oil and gold in the world. However, this indicator is expected to reach 8.7% in 2017 (World Bank, 2016).

Ghana is divided into ten different regions, illustrated in *Figure 1*, where the regions in the southern part of the country tend to be more developed than the northern regions. The capital is Accra, which is situated in the southern part of Ghana where most of the country's more economically developed cities are located. The northern region, as well as the upper east and the upper west, is mainly covered with savannah and is at large less developed and unexploited than the south. Ghana as a whole has achieved the extreme poverty reduction target under the Millennium Development Goals, inclusive of halving the proportion of population living below US\$1.25 per day between 1991 and 2006. The proportion of people suffering from undernourishment, including chronic hunger, has also reduced from 47.3% to 5% since 1990s (UN, 2015). However, the northern savannah zone, which counts for 40 % of Ghana's land area and 30% of its population of which 60% lives in rural areas, remains the least developed economy in the country (SADA, 2017).



Figure 1. Map of regions in Ghana, (GhanaWeb, n.d.)

3.2 Natural resources context

There are abundant natural resources in Ghana such as forestry and water, along with rich deposits of gold. Water availability, especially, in the Volta River, has contributed for hydropower to have a dominant share in electricity generation today. Still, there are other natural resources with large potential within the electricity generation sector, including solar potential and wind potential. In this section, the potential of electricity generation by several natural resources in Ghana is described.

3.2.1 Water resources

When it comes to the water supply and sanitary infrastructure, the access is insufficient, especially in rural areas of the country. According to the Joint Monitoring Program for Water Supply and Sanitation of UNICEF and WHO, 86 % of the population has access to an improved water system but only 18% of the population has access to water through house connections and only 14% has access to improved sanitation. Ghana also relies heavily on hydropower to meet its electricity demand, a technology which can be especially vulnerable to climate variabilities (Boadi and Owusu, 2017).

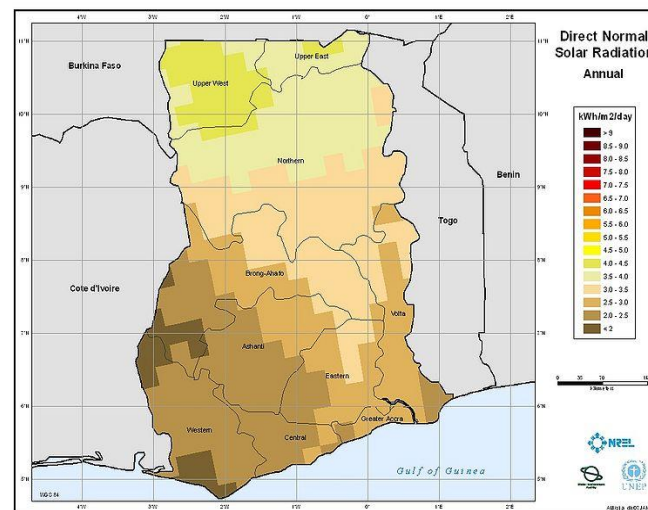
There is also a rising demand for water due to a growing industry and an expanding agricultural sector. Furthermore, Ghana's agricultural sector is highly sensitive to fluctuating water access since it is predominantly smallholder, traditional and rain-fed. The agriculture provides for over 90% of the food needs in the country as well as it accounts for 40% of its export earnings and contributes to 54% of Ghana's GDP. The main agricultural products in 2012 are cassava, yams and plantains while the main exported products in 2011 are cacao beans, refined sugar and cashew nuts (FAO, 2015, pp. 1-2).

According to Ghana's National Climate Change Adaptation Strategy (2012) Ghana's exposure to impacts from droughts and floods etc. define the country's vulnerability to climate change. Climate change in Ghana is expected to alter rainfall patterns, sea level rise, and increase the frequency of extreme weather conditions. These types of events are likely to affect the country's agriculture sector, hydroelectric power generation and water supply, both domestic and industrial (UNDP, 2012)

3.2.2 Energy resources

Ghana has, because of its access to the Volta River, a significant hydropower potential. The potential is currently being utilized by the three large hydropower plants Akosombo, Bui and Kpong. Beside these three plants there are at present 17 medium-sized and 22 small unexploited sites with capacities varying greatly between 15-100 MW. A combined total capacity of all the potential hydropower sites has been estimated to be around 800 MW (RECP).

Ghana's geographical location around the equatorial Sun Belt is a strategic location to exploit the abundant solar energy resources. According to the Africa-EU Renewable Energy Cooperation Programme (RECP), the average solar irradiation in Ghana varies between 4.5-6.0 kWh/m²/day and the amount of sunshine usually varies between 1800-3000 hours per annum. The highest irradiation intensities are found in the northern parts of the country which also generally has the most amount of sun hours (RECP,n.d.). Production of solar energy requires large land areas, both with Concentrated Solar Power (CSP) technology and with Solar photovoltaic (PV) farms which has a required area between 7-15 m² per kW (IRENA, 2012). Wasteland which is unsuitable for agricultural or residential use can be considered as construction sites. Most of the wasteland is located in the northern part of the country where the solar radiation is among the highest, which makes CSP and Solar PV promising technologies for this part of the country (Atsu et al, 2015).



<https://www.irena.org/DocumentDownloads/Publications/IRENA-ETSAP%20Tech%20Brief%20E11%20Solar%20PV.pdf>

Figure 3.2.2

Figure 2. Ghana Direct Solar Radiation Annual Map (NREL, 2005)

The potential of wind energy in Ghana is considered to be marginal. At 50 m over sea level the average wind speeds along the coast are between 4-6 m/s (RECP,n.d.). According to a study known as the Solar and Wind Energy Resource Assessment (SWERA) performed 2002-2005 by UNEP, there is over 400 km sq. area with good to excellent wind resources in Ghana. These resources could according to the same study be enough to support up to 2000 MW of wind power but it is estimated that only 500-600 GWh/year of electricity is possible to tap from the country's available wind resources (Mallet, K. 2012).

According to Energy Outlook for Ghana 2016, the delivered natural gas price from Western African Gas Pipeline (WAGP) was 8.75\$ per MMBtu, equal to 8.29 M\$ per PJ. Meanwhile, there is domestic reserve of natural gas being flown from Jubilee field through Atuabo in western Ghana. Atuabo is a gas processing plants, located in Western Region, Ghana. The gas processing plants has a big influence on Ghana's domestic power supply, as it contributes, in total, 500 MW in the total energy mix, also processes more than 180000 tonnes of liquified petroleum gas, which is about 70% of the total national requirement (Government of Ghana, 2017).

There are several gas fields in the country, and gas reserve can totally contribute to 5 trillion standard cubic feet per day. Besides Jubilee field, TEN (Tweneboa-Enyenra-Ntomme) field is expected to provide 63-70 mmscfd daily in the end of 2017. Other neighboring gas fields are Sankofa and Sankofa East fields, expecting to receive as much as 185 mmscfd daily in 2018, and this amount is twice the amount of supply from Jubilee field. Although this potential of natural gas supply is under development, the industrial use from these fields are expected to be after 2020. (Energy Commission, 2016)

3.3 The electricity system and electricity access

During the past ten years, the primary energy use in Ghana has increased with almost 50%. The usage by fuels has also varied in the recent years, though oil and wood have been the dominant fuels through the statistic period. In year of 2012, oil became the most used fuel for the primary energy generation and the use of oil exceeded for the first time the use of wood. In 2015 the energy production was thereby composed by oil, wood, natural gas and at last hydropower. From 2010 to 2015, the third biggest fuel varied between natural gas and hydropower. The demand and consumption of energy can be classified into as petroleum, biomass and electricity. The total energy supply 2015 was 9550 ktoe (Energy Commission, 2016).

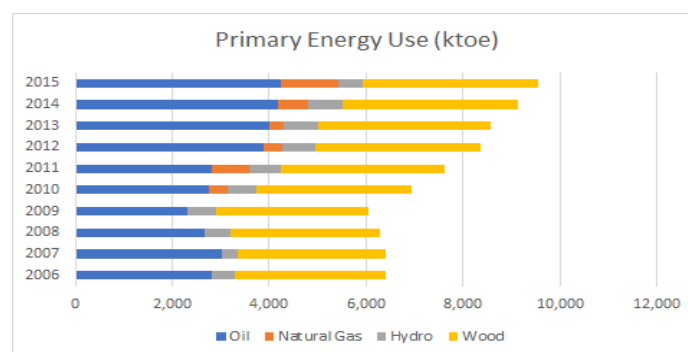


Figure 3. Primary energy use (Energy Commission, 2016)

Electricity consumption in Ghana 2015 was 8,646 GWh with a generation capacity of 3,656 MW. The electricity generation mix is composed of hydropower, thermal-power, mostly fueled by gas, oil and diesel, and the RET of Solar PV. Until 1998, hydropower had a large contribution to the electricity generation, but since then the contribution from thermal power has grown steadily. First in the 1980s Ghana suffered electricity outages because of droughts and reduced inflows into the river Volta; by that time the country was strictly dependent on hydropower for its electricity supply. Thermal power plants came thus first to complement hydropower. Also, compared with large hydropower projects thermal power plants have shorter construction period which is an advantage (Energy Commission, s.48, 2006). Today, the dominant energy sources in the electricity sector are hydropower and thermal power. In 2015, generation by hydropower and thermal power plants has very little difference, 5845 and 5644 GWh respectively (see figure 5), although there is more installed capacity of thermal power plants (see figure 6).

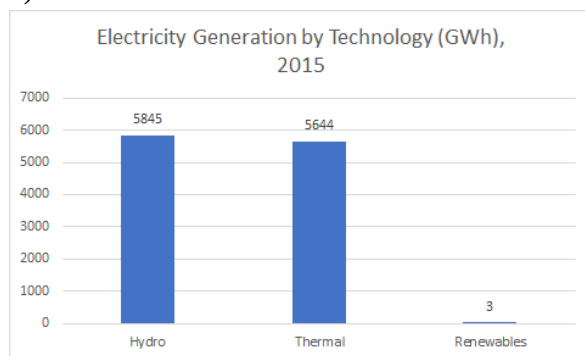


Figure 4. Electricity generated by technology

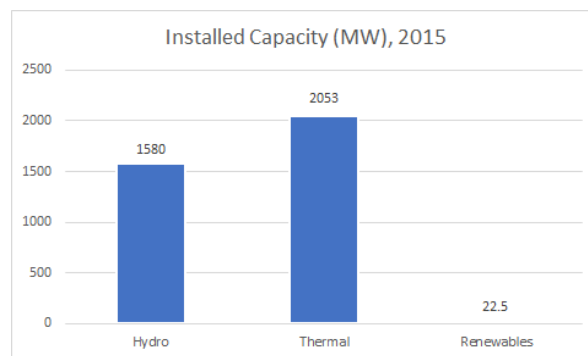


Figure 5. Installed capacity electricity generation

(Energy Commission, 2016)

Today about 64.1 % of the people in Ghana has access to electricity. But this number is much lower in the rural communities and the connection is often poor and unreliable. In 2012, it was estimated that 85% of the urban population had access to electricity while for rural population the figure was 41% (RECP,n.d.).

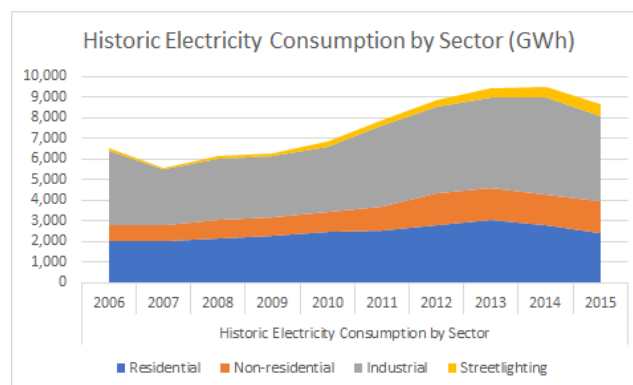


Figure 6. Historic electricity consumption (Energy Commission, 2016)

The third group in electricity generation sector is renewables. One example of renewable energy is solar power via photovoltaic, which is important for the rural areas of Ghana, according to the main energy report by Energy Commission (2006). Despite of the abundant potential of solar energy, the generation by this type of fuel is negligible in the whole energy production in Ghana (Energy Commission, 2006).

The most relevant groups of consumers of electricity are industry, residential, non-residential and street lighting. Industry consumes the most, representing nearly 50% of the total consumption (Energy Commission, 2016) with its major production areas in mining, light manufacturing and aluminum smelting (Commonwealth Network, n.d.).

3.4 Relevant development policies and initiatives

Since the mid-1990s, Ghana has launched the Ghana Poverty Reduction Strategies (GPRS) aiming at transforming Ghana into a middle-income country by 2020, inclusive that the country should achieve a GDP growth by 7-10% between 2003 and 2015 by agriculture sector, industrial sector and Information Communications Technology. One main challenge for the country's development is to supply reliable and sustainable energy required by a growing population and an expanding economy. For the development to be sustainable it is thereby crucial to also consider the impact on economy and environment from the energy use and production (Energy Commission, 2016, s.4).

According to The Strategic National Energy Plan (SNEP), Ghana's national goal for electricity access is 100 % universal electrification by 2020, where 30% of the rural areas are to be electrified by decentralized renewable energy sources (Energy Commission, 2006). Many plans and targets have been set to reach the national electrification goals, such as the Sustainable Energy for All Action Plan (SE4ALL) and the Renewable Energy Act 2011 which is a law that promotes the utilization of renewable energy sources (Energy Commission of Ghana, 2011).

The Renewable Energy Masterplan (REMP) for Ghana was published in 2016 resulting from a joint effort of the Ministry of Power, the Energy Commission, the National Development Planning Commission and the Energy Center of Ghana. One of the objectives of the REMP is to set clear targets for the development of RES in Ghana as well as define necessary actions and strategies to achieve the targets. The targets involve a substantial increase of the electricity generation share by solar and wind technologies between 2016 and 2030 as well as increasing share of standalone and mini-grid electricity distribution solutions with solar and wind technologies (UNDP, 2016).

Table 1. Renewable Energy Targets from the Renewable Energy Masterplan 2016 (Renewable energy master plan, 2016)

Intervention	Solar PV	MG Solar PV	SA Solar PV	Wind
Capacity 2020 (MW)	150	50	47	225
Capacity 2025 (MW)	225	100	129	375
Capacity 2030 (MW)	300	200	210	500

4. Methodology

The following section is dedicated to describing the methodology followed in the development of this study. In essence, it results from the combination of a qualitative and quantitative analysis. On one hand, interactions between CLEW systems are identified and selected for further investigation. On the other hand, the latter is achieved through the development of models of the electricity system from which quantitative outputs are derived. The research is mainly based on national official data, however, whenever data proves to be insufficient, assumptions will be made to further the analysis.

Primary steps of the study are as follows:

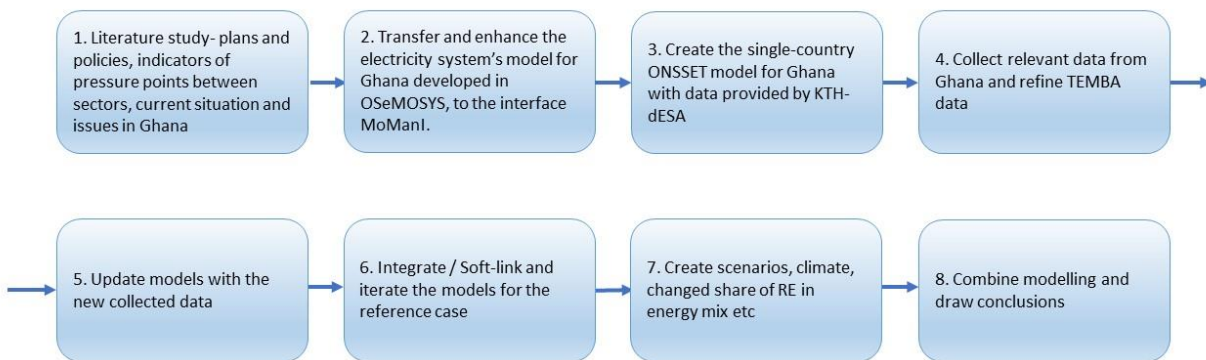


Figure 7. Primary steps of the study

4.1 The Climate, Land use, Energy and Water strategies

Valuable resources, such as land, energy and water are being exploited in such a way that it is changing the climate. Whilst, the systems providing these resources are themselves highly dependent on the climate and therefore vulnerable to climate change. This means that it is of great importance to manage the resources in an efficient and sustainable way, to minimize the impact on the environment and the reduce of limit the GHG emissions connected with the exploitation of these resources.

Furthermore, these resource systems are highly interlinked and their management would benefit if planned following an integrated perspective. The Climate, Land use and Water strategies is a framework for the efficient management of these valuable resources (Howells et al, 2013). This approach focuses on identifying the different points where these systems interact such as pressure points and trade-offs, and assess these so to identify and explore potential cross-sectoral opportunities and synergies. The assessment of relevant interactions is achieved through the development of sectoral models that are then soft-linked. Input and/or output data used in or retrieved from the modelling exercises can be exchanged between the different modules, providing an integrated and quantitative picture of the use of resources for different potential futures. These are defined based on national policies and strategies, or may result from cause-effect investigation of sectoral measures. The idea of the assessment is then to let the output data from one module form as input data in the other two modules. It is then possible to analyze the interlinkages between the modules such as the need of water for land-use and the energy sector, the

requirement of land use for water and energy infrastructure, as well as the energy need for land use and water supply (Howells et al., 2013).

A relevant example for the situation in Ghana are the links between energy, water and climate. The country's energy sector is today greatly dependent on water for hydroelectric power generation, which makes the energy sector vulnerable to water related changes such as changes in precipitation and evaporation, which are projected effects of climate change (UNDP, 2012). In this case study of Ghana, the CLEWs framework is applied by studying interlinkages between the energy sector, the climate and the water sector. This is done by studying the GHG emissions from the electricity generation mix and the effects of climate change on hydropower production, and consequent implications to the development of the electricity system.

4.2 Energy modeling tools

The project focuses greatly on the analysis of the energy sector, in particular, the electricity system. This requires the development of a model to represent the electricity system of Ghana. This will be achieved using the open-source long-term energy systems software OSeMOSYS (Howells et al, 2011).

OSeMOSYS, developed by KTH-dESA and other partners, operates with the objective to present the lowest net value cost for the energy system that meets the demand for energy and energy services. The model is used for long-term energy planning, for exploring different developing scenarios and for visualizing the optimal future energy mix (Howells et al., 2011). Additionally, and in order to refine the electricity sector analysis, the geospatial electrification tool ONSSET (Mentis et al, 2016) will be deployed to refine the representation of the electricity system in OSeMOSYS, taking into consideration the achievement of 100% electrification access by 2030. ONSSET, developed by KTH-dESA, is used for the achievement of electricity goals, and should present the optimal electricity distribution mix for electricity access of non-electrified households, including stand alone, mini-grid and national grid systems (Mentis et al., 2016). The end-year is currently defined for 2030 to inform on the achievement of 2030 Agenda where the SDG 7 is included (more specifically target 7.1: to ensure universal access to reliable, affordable and modern energy services by 2030), however the end-year can be adjusted by the user. Both models, ONSSET and OSeMOSYS, incorporate national data as much as possible. Assumptions were made whenever national data was not available or was inconsistent. The Model Management Infrastructure (MoManI) is an interface to facilitate the use of OSeMOSYS, and was used for the data input of different scenarios. Results generated by OSeMOSYS and ONSSET provide the most economically optimal solution for the electrification of the country considering different technological options and electricity system costs. More information on the energy system tools can be found in section 4.3.

4.3 Integration of tools in the case study

This section presents a description of how the tools indicated in the previous section were soft-linked for this case study of Ghana, as well as what was taken into consideration in the harmonization of parameter inputs.

4.3.1 Energy system optimization - OSeMOSYS

To determine the optimum electricity generation mix for the grid connected technologies in Ghana, the OSeMOSYS modelling tool was used. The model finds the most cost-effective solution for the energy mix depending on the technologies and fuels represented. The interface MoManI was used in the deployment of OSeMOSYS. The objective of the model is to evaluate the most cost optimal generation mix that can fulfill the demand of 100% electrification up till tier 3 in 2030 (696 kWh/household/year) and tier 5 in 2050 (2195 kWh/household/year).

General data for the electricity system of Ghana was obtained from The Electricity Model Base for Africa (TEMBA) (Taliotis et al, 2015; Pappis, 2016) and used as a starting point for model development of the baseline case. To fulfill the need of creating scenarios, diverse sets and parameters need to be defined in MoManI. First of all, the present used fuels and technologies within energy sector should be presented. In this case they are composed by both renewable fuels such as water, solar and wind, and also fossil fuels such as oil, diesel and natural gas. Energy demand is often dependent on time, for an example it may differ over the seasons. In TEMBA, time slices were divided in summer day, summer night, winter day and winter night.

Also, the operation of power plants incurs fixed and variable costs, and is defined by their efficiency. Parameters for capital cost, fixed cost and variable cost together with investment capacity represent the cost of energy technologies. Another crucial factor is efficiency of these technologies. In order to present how effective, the technologies are, there are parameters for availability and capacity, e.g. availability factor, capacity factor and its maximum and minimum capacity for a certain year. The efficiency for thermal power plants are represented using input and output activity ratios. To be able to make an analysis of the effects on the environment and climate change, emissions should be taken into consideration too. Therefore, there is an activity ratio of emission in TEMBA, and this parameter represents GHG emissions in carbon dioxide-equivalent. (Howells et al., 2011)

There are time-independent factors that also have impact on the performance of energy technologies and help get an understanding of them. These are for example operational life of the power plants, and the total energy production by one power plant for one year. (Howells et al., 2011)

Because TEMBA represents data that is less specific and accurate for Ghana, some of the relevant parameters mentioned above, such as capacity factor and fuel price are expected to be updated in later steps of the study with newly collected data.

4.3.1.1 Reference energy system (RefES)

The RefES is a graphical abstraction of the electricity system of Ghana that will be modelled with OSeMOSYS via MoManI. The abstraction the different technologies in the system, such as power plants and transmissions, are indicated by blocks and the energy and energy services, such as fuels and electricity, are represented by lines and are connecting the different technologies. The RefES is restricted in the way that it only represents the import and extraction of energy generating fuels in Ghana, thus trade links with other countries are not taken into account.

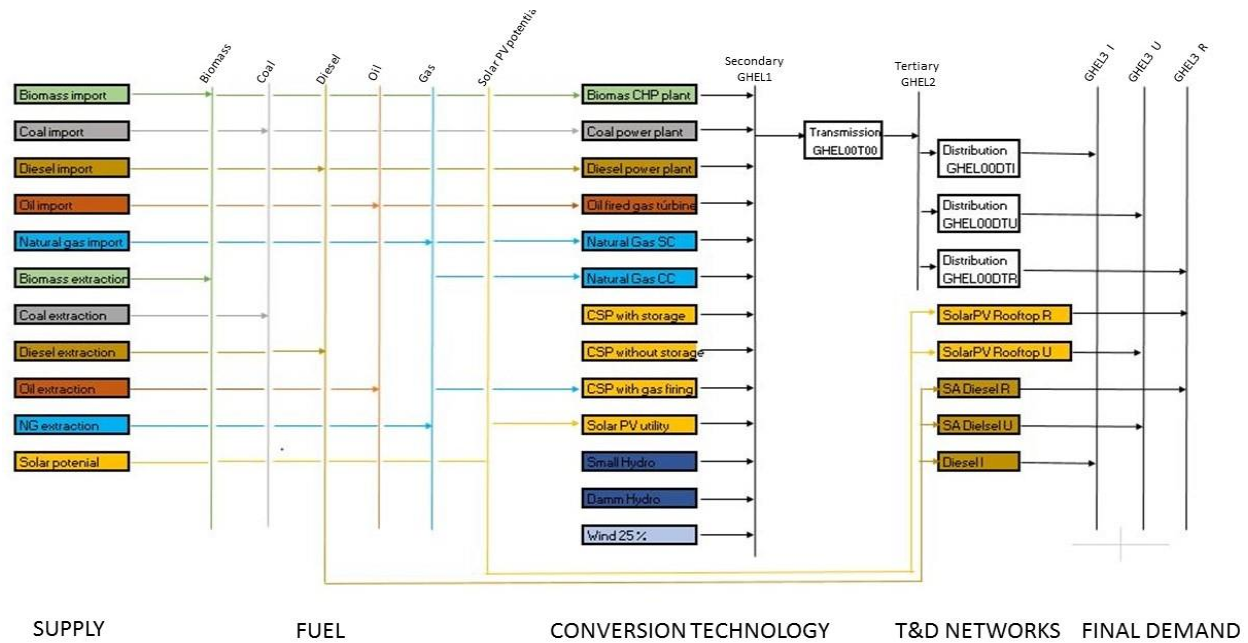


Figure 8. Reference energy system

4.3.2 Household electricity access with ONSSET

The toolkit ONSSET is being used for the residential electrification analysis. The model creates an optimized spatial electrification plan for Ghana, based on detailed geographical information (GIS) and costs for the different technologies included in the system. The objective of this analysis is the achievement of 100% residential electrification (up till tier 3 which equals 695 kWh/household/year) in Ghana by 2030, with the base year set as 2015 (UN-DESA, n.d.). The ONSSET model informs on the optimal split between the different electricity supply solutions for the household demand without access to electricity in the reference year of 2014. The result will show the share between grid and off-grid solutions. The off-grid solutions consist of mini-grid and standalone systems such as solar PV, diesel generator and mini-hydro and -wind.

The results will vary depending on the chosen tier of electrification, which represents the annual electricity consumption per household for non-electrified households in the reference year of the analysis. In the base year, no distribution technologies are taken into consideration, hence the model assumes that all electrified households in that year has access to the national grid. This assumption is rather accurate in the case of Ghana where the grid is relatively well developed while standalone solutions are not. Furthermore, the chosen tier of electrification is only applied to determine the demand of the non-electrified households and does not take into consideration the households with electricity access in the base year, they are thereby assumed to keep their current electrification supply solution.

4.3.3 Model integration

The OSeMOSYS model and the ONSSET model have different structure and follow a different methodological approach, and must therefore be connected with specific shared parameters. In the integration, data will flow between the models in the following way: first the OSeMOSYS model will be used to calculate the cost of generating electricity by the grid. In this step, no part of the demand will be

satisfied by other technologies than the grid connected ones. This cost will be obtained by calculations externally, using outputs from OSeMOSYS that are related to the electricity generating technologies that supply the grid. This parameter, named in ONSSET as Grid Cost, will then be used as an input in the ONSSET model, to optimize the spatial electrification optimal split for the country. The results from ONSSET, related to the electricity demand supplied by off-grid solutions, are then used to modify the share of the demand electrified by off-grid technology options in OSeMOSYS. This is done by applying limits of generation to the related technologies in the OSeMOSYS model. After running the OSeMOSYS model again with the new alterations, re-calculation of the Grid Cost is necessary. The procedure is then iterated until the values correlate and the grid cost does not differ more than 10% and thereby does not affect greatly the combination of off-grid technologies.

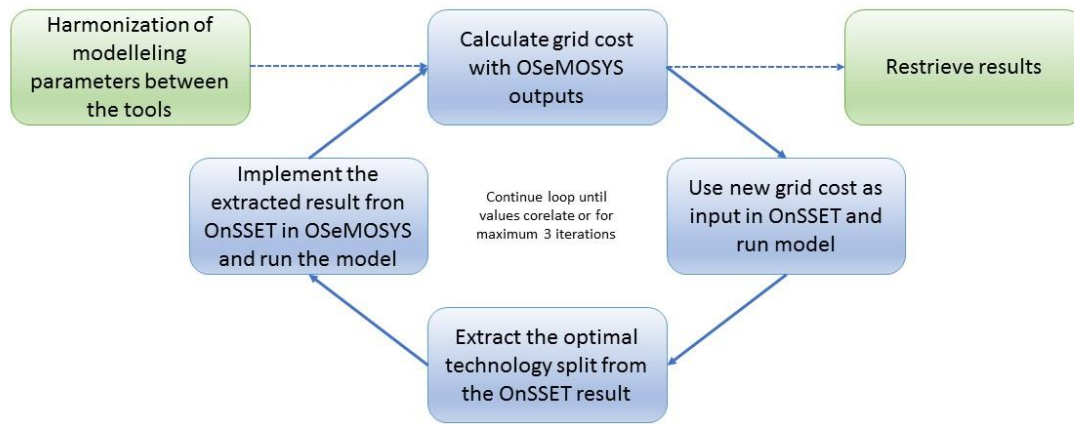


Figure 9. Integration between ONSSET and OSeMOSYS model

4.4 Model development

In the first stages of modeling the electricity system in Ghana, general data from a multi-country model of Africa (excluding island states) called TEMBA, is used as inputs in MoManI. To develop the models used in the study, both country specific data and assumptions are used to replace the values from TEMBA when suitable. The process of updating the data and developing the models is described in the following section.

4.4.1 TEMBA

In the first stages of modelling, data based on the whole of Africa, also called TEMBA (Taliotis et al, 2015; Pappis, 2016) is used as inputs in MoManI. This means that though this general data should be the foundation, improved data with specific information about the energy system in Ghana should be used to update the model to receive the most realistic result of the electricity generation. The specific data for Ghana is obtained from reliable documents of energy related authorities, such as Energy Commission of Ghana and Savanna Accelerated Development Authority.

4.4.2 Data gathering and main assumptions

An important step of the work in Ghana was the collection of country specific energy related data. On site, the chances of getting access to relevant and specific information is higher since personal interactions and physical meetings are possible on site. The specific data for Ghana is obtained from reliable documents of energy related authorities, such as the Energy Commission of Ghana.

Assumptions are used when data does not exist or is inaccessible. The assumptions used in this study concerns new small hydropower plants and their expected construction time frame; an assumption is used to decide when the plants are expected to start to generate electricity. When modeling the two different climate scenarios, dry and humid, these small new hydropower plants are assumed to have the same capacity factor. Assumptions are also used when projecting the future fuel prices.

4.4.3 Model update

The collected data and the assumptions are then used to update the models used in the case study. The OSeMOSYS model is first updated with current fuel prices that are specified to Ghana. Furthermore, newly built power plants that has not been included in the data set TEMBA, and power plants that are under construction or planned to be constructed during the time frame of the projection, are added to the model and can thereby be considered for the electricity generation mix modeled. In this case, they are small hydropower plants with limited capacities. The OSeMOSYS model is also updated with the current national targets of the share of renewable energy in the energy mix. This is done by forcing the model to invest in a certain amount of RE capacity composed by Solar PV and Wind technologies. This is done without affecting the actual electricity production, the model is not forced to produce electricity with these technologies if it is not the most cost optimal solution.

4.5 Scenarios

To determine the impact of different assumptions and actions, different scenarios can be developed and implemented in the completed model framework. In the case study of Ghana, scenarios related to climate change will be developed to evaluate how the occurrence of a changing climate may affect the system.

4.5.1 Climate Change Scenarios

Climate change has an impact on many sectors in Ghana, including the energy and agricultural sectors, because of their reliance on nature resources. The historical statistics of climate in Ghana, from 1961 to 2000, show an increase of mean temperature across this period while a decrease of precipitation (UNDP, 2012). Since more than 50% of Ghana's electricity generation comes from hydropower today, the electricity sector is very vulnerable to climate change, especially change in rainfall patterns (UNDP, 2012).

The electricity system in the case study is modelled with three different climate scenarios. The first one is the Business as usual (BAU) scenario where the climate is assumed to be similar to the recent past during the time frame of the projection. The other two scenarios represent a climate that changes and gets drier in the future and a climate that becomes more humid in the future. The change is cumulative and might vary across the years of the projection.

5. Results

Results of the most cost optimal energy mix and distribution solutions are presented in this section. The results represent different possible futures for Ghana's electricity system, including Business as usual, a future with more humid climate, and a scenario representing a drier future than today. The results presented in this section have been generated through an integrated process between the modeling tools OSeMOSYS and ONSSET.

5.1 Business as Usual

In addition to the power plants which already exist, installed capacity targets and plans of new power plants are included in the model. Results of annual production by technology, obtained from OSeMOSYS, provide an indication of the configuration of the electricity generation mix during the period 2010 to 2050, in consideration of the most cost optimal solution.

The results show that a combination of both renewable and fossil fuel-based electricity generation, including CSP, natural gas, hydro, solar PV, coal and wind together gives the electricity generation mix with the lowest cost in the future (see figure 10). In the result, CSP and natural gas are the two dominating technologies, generating 357 PJ and 133 PJ respectively in 2050. The electricity production by CSP is more than 50% of the total production in the end year. Solar PV and wind are the other two renewable energy technologies generating electricity in the result produced by the models. Although the contribution by wind is far greater than that by Solar PV, the generation is still relatively low compared to the generation by CSP. Hydropower is the dominant source for electricity generation in Ghana today, and there are plans to expand this sector further, mostly in form of small hydropower plants. In the BAU projection, there is no clear trend of hydropower production increasing, rather it decreases slightly during the last ten years of the projection.

Natural gas and coal are two main fossil sources for electricity production in the BAU projection. Together they compose 27.6 % of the total generation in 2050, with the usage of natural gas being much higher than coal. This trend is a result of domestic natural gas reserves, which give a relatively low fuel price compared to other fossil fuels, such as oil and diesel. According to the statistics, the domestic natural gas price was 8.84 US\$ per MMBtu, whereas the LCO price was 60 US\$ per barrel in 2015, which corresponds to 8.3 respective 10.2 million US\$ per PJ. (Energy Commission, 2016).

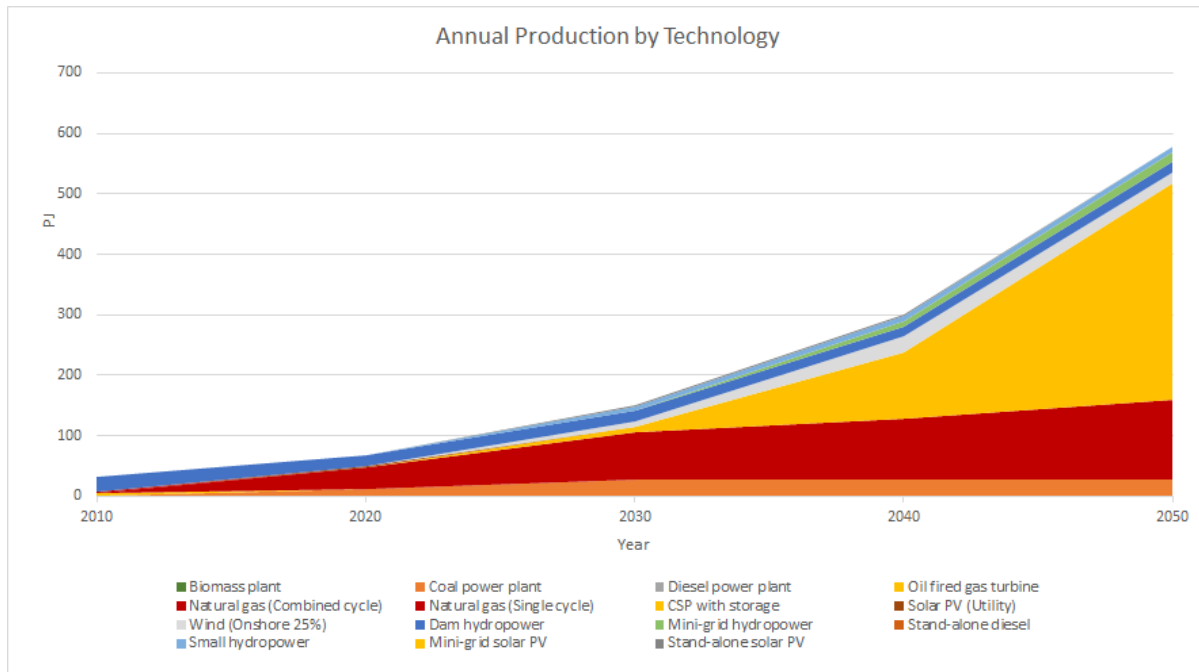


Figure 10. Annual production by technology for BAU scenario

The results for annual emissions shows the annual CO₂ emissions from the total electricity generation in Ghana until 2050. As can be seen in figure 11 the amount of emissions is increasing clearly and drastically the first years of the projection, but the rate of increase slows after 2030.

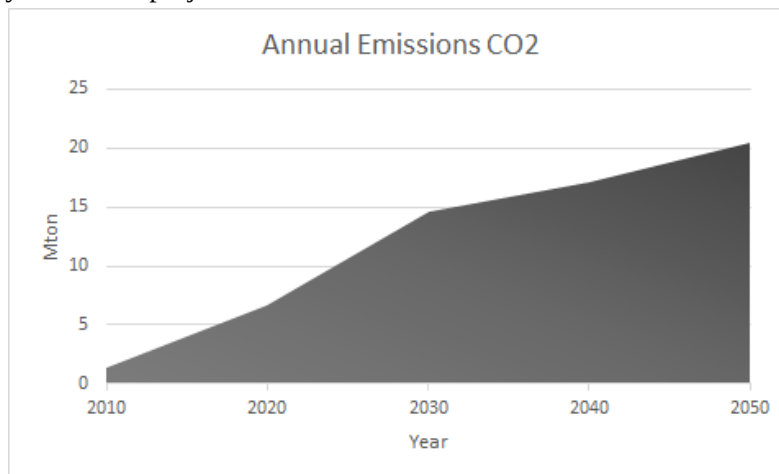


Figure 11. Annual emissions for BAU scenario

The results from ONSSET show the optimal electricity distribution split in 2030, where the grid price is based on the electricity generation mix derived from results from the OSeMOSYS model (see table 2). An increase in the deployment of standalone and mini grid solutions, especially for standalone Solar PV and mini grid hydro, is verified for this scenario. Still the majority of newly electrified, about 94%, are connected by the grid distribution technology..

Table 2. The summarized results for full electrification of Ghana by 2030, BAU

	Population	New connections	Capacity (kW)	Investments (million USD)
grid	36268015	18655288	640723	5789.50
sa_diesel	5517	5517	357	0.54
sa_pv	931115	931115	97657	537.12
mg_diesel	0	0	0	0.00
mg_pv	21653	21653	2301	13.79
mg_wind	0	0	0	0.00
mg_hydro	257251	257251	11662	126.21
Total	37483553	19870826	752702	6467.16

The maps presented in **Figure 12**, were retrieved from ONSSET and show the geospatial results for the electricity cost and the distribution of technological options to supply households in 2030, which did not have access to electricity in 2014, taken into consideration population growth. The first map on the left shows the electricity cost and how it varies across the country. The map on the right shows the technological split between grid, standalone (SA)diesel, SA PV, mini-grid (MG) diesel, MG PV, MGWind and MG hydro (see Figure 12). From this result, it is clear that the standalone solutions (presented in the figure in yellow for SA PV and light-yellow for SA Diesel) will be more cost optimal in the northern parts of the country where the electricity grid is less developed and a larger share of the population lacks access to electricity, as compared with southern regions. Comparing the two maps, the results also shows that these areas will have a higher electricity cost than the areas supplied by the grid.

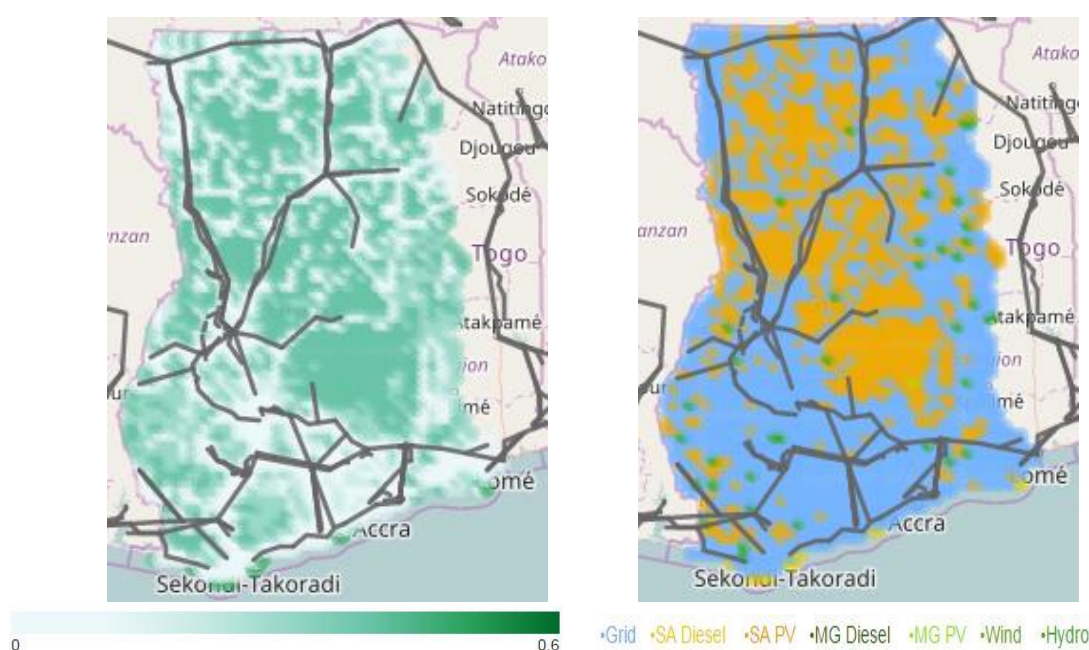


Figure 12. Electricity cost (left) and optimal technological split (right) for the BaU scenario.

5.1.1 Climate Dry

Same type of results as for the BAU scenario is presented in this section. In the projection of the dry climate, CSP and natural gas still dominate the production in the electricity generation mix of Ghana in the future. Wind power, hydropower and coal are other important technologies, even though the contribution of these are small in comparison to the contribution of CSP and Natural gas.

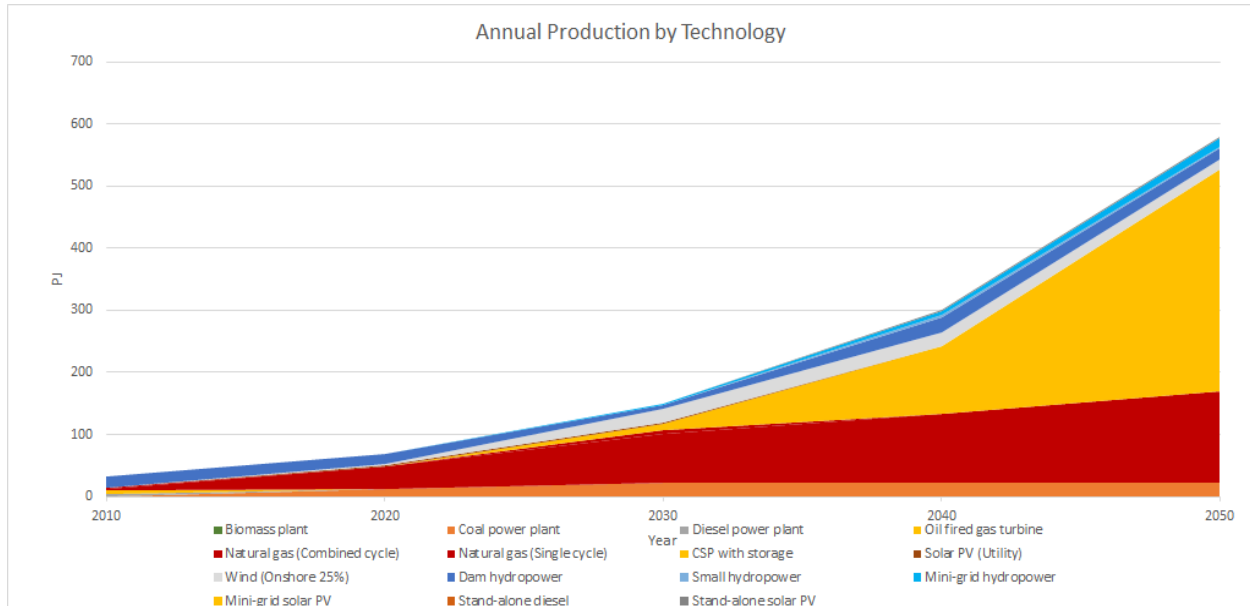


Figure 13. Annual production by technology for dry scenario

The CO₂ emissions from the electricity generation is shown in the graph below. There is a clear trend of increasing emissions during the whole timeframe, but the increase is most extreme between 2020 and 2030, where the share of natural gas is increasing to be the dominating technology of the generation mix. After 2030 the contribution of CSP grows steadily, and the emissions subside.

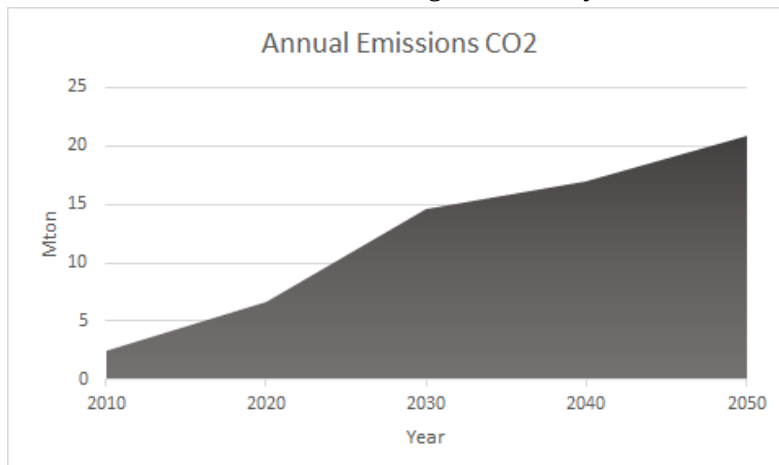


Figure 14. Annual Emissions for dry scenario

The result from ONSSET shows that a smaller share of the new connection is electrified by the grid as a distribution technology than in the BAU scenario. The difference is very small but clearly a consequence of the higher grid price generated by the OSeMOSYS model working as an input in ONSSET.

Table 3. The summarized results for full electrification of Ghana by 2030, Dry scenario

	Population	New connections	Capacity (kW)	Investments (million USD)
grid	36016274	18403547	632077	5653.39
sa_diesel	5517	5517	357	0.54
sa_pv	1108056	1108056	116305	639.68
mg_diesel	0	0	0	0.00
mg_pv	21653	21653	2301	13.79
mg_wind	0	0	0	0.00
mg_hydro	332051	332051	15053	156.57
Total	37483553	19870826	766095	6463.97

5.1.2 Climate Humid

The result for production by technology for the humid climate scenario is similar to the other two scenarios. The main difference is the amount of electricity generation by hydropower plants. The other main technologies contributing to the generation mix are CSP and Natural gas. Coal and wind power only contribute marginally to the production in this scenario (see figure 15).

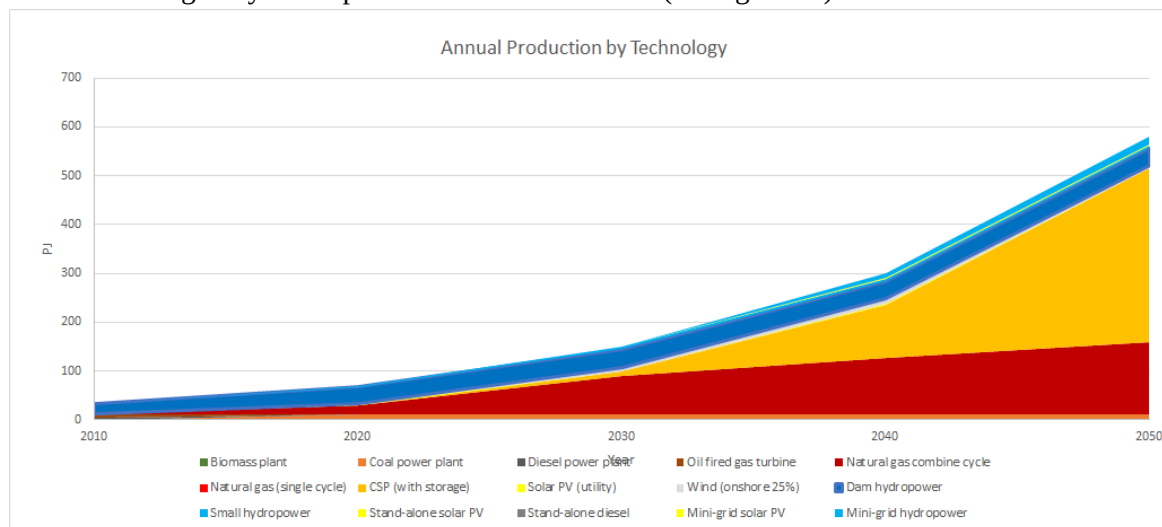


Figure 15. Annual production of electricity by technology for humid scenario

The results for annual emissions shows the annual CO_{2,eq} emissions from the total electricity generation in Ghana until 2050 for the humid scenario. The amount of emissions is increasing steadily in the first years of the projection, and more drastically between 2020 and 2030 where natural gas contributes to more than

half of the production share in the generation mix. After 2030, CSP is introduced in the mix, and the rate of increase of emissions starts to slow after this. This emission value is lowest compared to the other two scenarios.

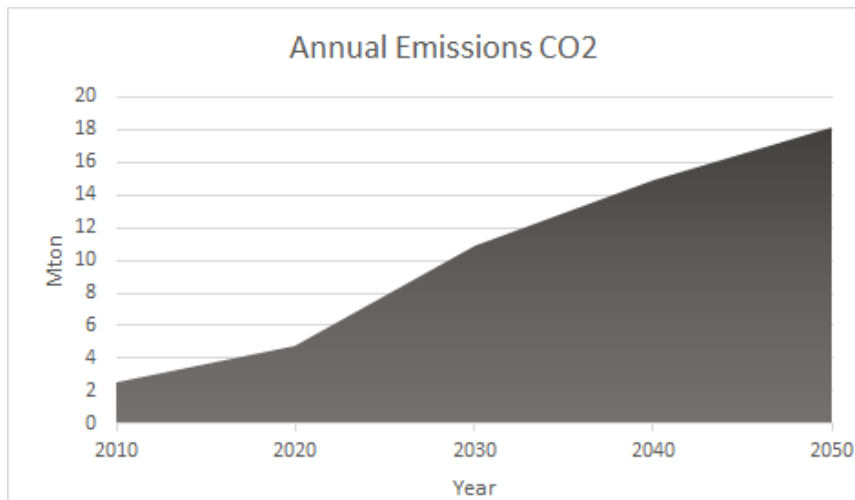


Figure 16. Annual Emissions for humid scenario

The grid price working as an input in ONSSET is lower than in the other two scenarios, which makes the grid a more profitable distribution technology solution as shown in the result table obtained from ONSSET below (see table 4). The fact that the new connections to the grid are increasing compared to the BAU and Dry scenario, means that new connections by other distribution technologies decreases.

Table 4. The summarized ONSSET results for full electrification of Ghana by 2030 in the “Humid Scenario”.

	Population	New connections	Capacity (kW)	Investments (million USD)
grid	36471815	18859088	647723	5886.49
sa_diesel	5517	5517	357	0.54
sa_pv	869751	869751	91246	501.86
mg_diesel	0	0	0	0.00
mg_pv	21653	21653	2301	13.79
mg_wind	0	0	0	0.00
mg_hydro	114815	114815	5205	62.18
Total	37483553	19870826	746833	6464.86

The maps shown below of the electrification results represent the electricity cost at each point and the spread of distribution technologies based on geospatial data. The maps representing the humid scenario, may look very similar to the maps retrieved for the other two scenarios, but as the table above shows, there is a small increase of new grid connections in the humid scenario at 2030; 94.9 % compared to 93.9% in BAU and 92.6% in Dry. The larger amount of grid connection also affects the technology split as a whole.

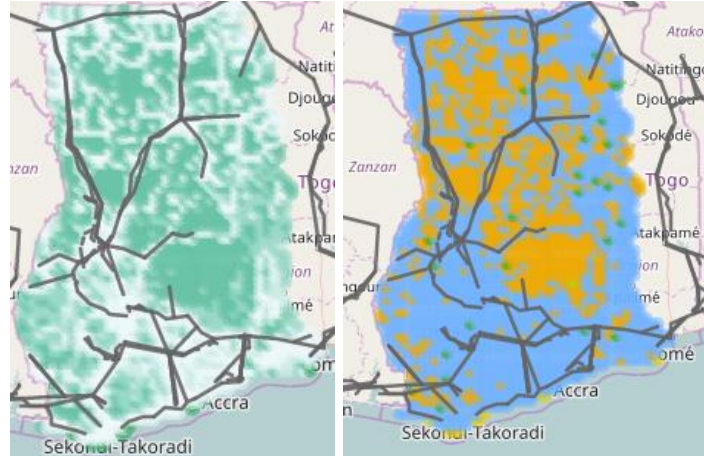


Figure 17. Electricity cost (left) and optimal technology split for the “Humid” scenario.

5.2 Comparison between scenarios

There are clear differences of performance and production in different climate scenarios. During humid climate, the electricity production by hydropower is much higher, which indicates the fact that it will be the most economically optimal. Also, the production has almost reached its peak very early, around year 2020. In the contrary, the production by hydropower is not only lower, but also unstable with much variation during the dry state within shorter periods. This could be a result of its capacity factors, as they do vary as well.

In BAU scenarios, the production and performance is relatively stable, there are not as many sharp changes as in the other two scenarios. The overall production in BAU is between the wet and dry states, but during some shorter periods production of the dry scenario exceeds BAU.

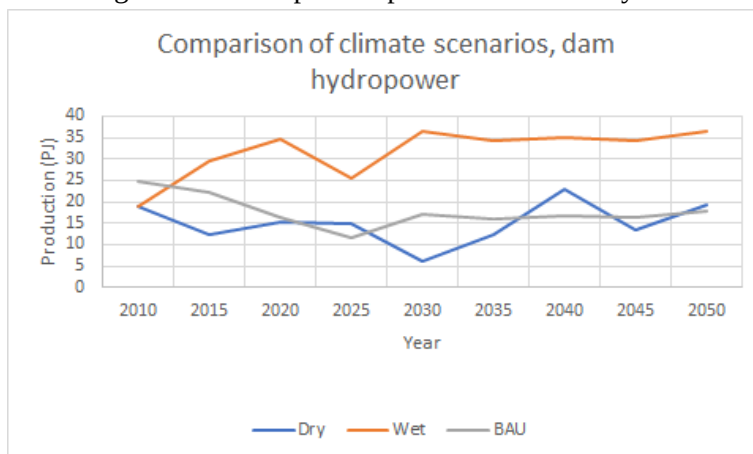


Figure 18. Comparison of climate scenarios, dam hydropower

In order to be able to compare the emissions of different climate scenarios, the emissions are presented as a ratio between annual emission in million tonnes and the annual total production in petajoules (PJ). The result shows in general that in the wet scenario the emission ratio is lowest and the dry scenario has the highest ratio. Between 2010 and 2025, BAU has 7 lower emissions than in the dry scenario, but sometime

around 2030 the emissions begin to have a very similar trend for these two scenarios. In the end of the modelling period, 2040 - 2050, the difference between the emissions of wet and other two scenarios decreases.

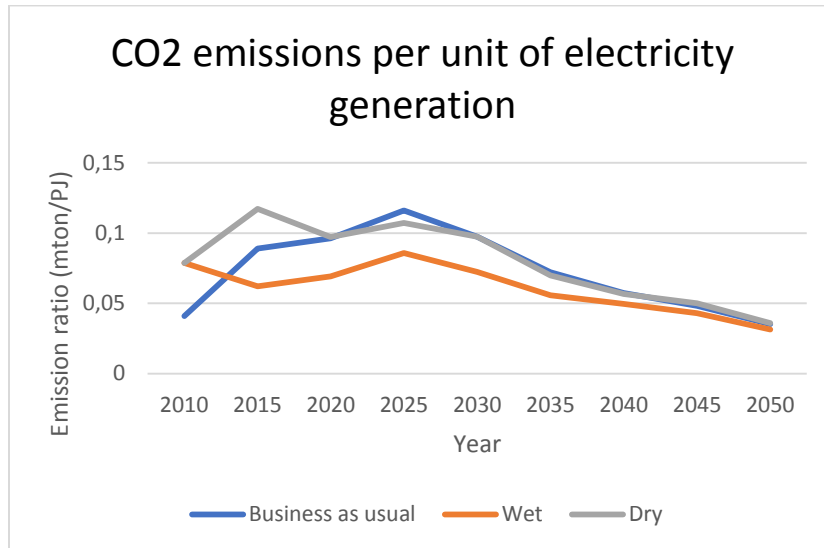


Figure 19. Emission per Production

6. Discussion

The following section is dedicated to discussing the result of the study. The analysis focus on how realistic the result is considering the actual production potential by different sources in Ghana, and the country's energy related targets and goals, affecting the development of the electricity system. The model's choice of different energy combinations in the generation mix is also analyzed in order to understand what is making the mix cost optimal. Furthermore, the section raises the issues surrounding hydroelectric power and its vulnerability to climate change as well as Ghana's achievement of the SDG 7.1.

6.1 Choice of technology

The models consider that a combination of CSP and natural gas together with other renewable energy technologies is the most optimal electricity generation mix for Ghana in the future. This energy mix is not only cost optimal, but also considered environmental friendly as in 2050, 72 % of the total production will be obtained from renewable energy sources. Although this is positive for sustainable development, the very high share of CSP coming into the mix is worth discussing, as it is a newly developing solar technology and information of its actual performance may be uncertain.

Although there are three CSP technologies working as inputs in the model, CSP without storage, CSP with storage and CSP with gas co-firing, the dominant share comes from CSP with storage. This is likely

a result of the technology's high performance, presented by both its capacity factor and its variable cost per unit of produced electricity. On average, the CSP technologies have high capacity factors, unlike solar PV. In the OSeMOSYS model, time slices are divided into summer night, summer day, winter night and winter day which makes it possible to take into consideration the technology's differences in performance over the whole day. The capacity factor for CSP varies between 55% to 92%. For CSP with storage and with gas firing, there is also a possibility of storing the heat energy which makes it possible to generate electricity by night. CSP with storage has a capacity factor as high as 70% during the day and 55% during the night, compared with 70 % during the day and 0% during the night for CSP without storage and 92% for CSP with gas firing regardless of the time a day. Variable costs of CSP with storage is at average 0.68 mUSD per PJ produced energy which is much lower than the cost of 4.56 mUSD per PJ for the CSP technology with gas firing (Pappis, 2016).

CSP has an obvious advantage over solar PV, as its capacity is much higher and the potential of storage is greater. Compared with other technologies in the models, CSP has the advantage of a relatively low variable cost, since no other fuels other than sun's heat is needed, and fuel price is often a constraint limiting the usage of a certain technology.

Natural gas combined cycle is the second largest technology providing electricity in the future. Natural gas is already supporting many thermal power plants in the country presently. The statistic shows that Ghana uses both domestic gas from Atuabo and imported gas from Nigeria. Compared with other fossil fuels such as oil and diesel, natural gas is relatively cheap. In 2015, the gas price expressed in mUS\$ per PJ was 8.34, whereas the oil price and diesel price were 10 and 15. Also, the domestic reserve could ensure cheap supply of natural gas for the electricity system in Ghana. Furthermore, natural gas combined cycle is a well established and mature technology and its capacity factor is as high as 93.5%, its variable cost is at 0.55 - 0.66 \$/GJ. Thus, the technologies that use natural gas should be able to provide great amounts of electricity without large losses.

6.1.1 CSP vs Solar PV technologies

An interesting aspect shown in the result is the large amount of electricity generated with Concentrated solar power (CSP) in the future compared with the amount generated by Photovoltaic solar power (PV). Today Solar PV is the preferred energy solution over CSP for energy investors, due to its lower energy price. It is thereby relevant to discuss why CSP potentially could be more economically optimal in the future, and if the result provided by the model is realistic.

The CSP system concentrates the solar radiation to heat a liquid substance, that are then used to generate electricity through a heat engine that then drives an electric generator (Green, 2012).. The photovoltaic solar panels on the other hand are not using the heat from the sun to generate the electricity, but the actual sunlight. This makes the technology unusable in the absence of direct sunlight. The Solar PV technologies generates electricity directly, which is hard to store with the current available technologies (batteries), especially at large power levels (Green, 2012). Due to the CSP systems ability to store electricity with the use of thermal energy storage technologies (TES), the technology is more favorable for the model to choose, compared to the Solar PV technologies. This, even though, the CSP technology has higher investment costs as well as higher variable costs during the production.

The electricity demand in Ghana is projected to increase greatly from the base year until 2050. From this follows that the electricity generation mix naturally will undergo great changes during these years. Due to the TES, CSP system is more attractive for large scale power generation, which is required due to the high electricity demand. Since the CSP technology can produce excess energy during the day that can be used during periods where there is an absence of radiation from the sun, the technology will still figure in as an option for the model to choose in the electricity generation mix, while the Solar PV technologies will be unable to produce during these periods. One can thereby argue that the result could be presenting a realistic future electricity generation mix, even though Solar PV today is the more frequently used technology of the two solar systems that are available options in the model.

6.1.2 Is the result realistic?

It is obviously difficult to project how the electricity system will look in over thirty years. A lot of parameters determining the outcome might change in ways that are not taken into consideration in this study. It is therefore hard to determine how reliable the result generated by the model is. Since Ghana is lacking great oil reserves and since the country's hydropower sector already is largely developed and the remaining potential is limited (see section 3.2.2 for precise numbers), it is realistic that the increased generation of electricity will have to come from other technologies.

In the model, there are no targets or constraints forcing it to choose a larger share of RE-technologies, but the result still shows that CSP and hydropower has some of the largest contributions to the generation mix within RE-technology in the future. It can be discussed whether or not CSP can contribute to the electricity production in Ghana on a large scale. It is possibly unrealistic since the technology is rarely used presently, however it does in any case show the importance of solar energy for Ghana to meet its increasing electricity demand in the future. The other big contributor to the generation mix is natural gas. This projection can be considered to be realistic since Ghana has access to reserves of this fuel, which make the technology cheaper for the country to use.

Ideally the generation mix should consist of a variety of technologies to secure a reliable and constant production of electricity. To rely too heavily on one source, especially one like CSP that is dependent on an inconsistent fuel like sunlight, is not wise since it makes the production vulnerable to outside changes. Therefore, the produced result should not be viewed as completely realistic, due to the projected future electricity productions large reliance of CSP.

6.1.2 Sustainable development goal 7

To answer the second research question, it is necessary to analyze if the result obtained by the model will contribute to the achievement of the sustainable development goal 7.1, and, if that is the case, if this achievement is due to the contribution of investments in RE technologies in the generation mix.

The affordability of the electricity generation obtained by the model is most easily analyzed by studying the grid price used as an input in the ONSSET model. The grid price presented in BAU scenarios is 0.0696 USD/kWh. This can be considered to be a reasonably low price and thereby contributes to affordable electricity access for the people in Ghana. The result that the grid price remains reasonably low, could be a direct result of the large contribution of RE technologies in the generation mix. These

technologies may have large investment costs, but the costs of generating electricity when the power plant is installed is usually lower than corresponding fossil fuel dependent technologies. However, the grid price alone is not enough to analyze the affordability of the electricity generated in the country, since the total cost also is highly dependent on the cost of expanding the grid and thereby updating its capacity.

The cost of providing electricity in a low populated, isolated location that is far from roads and the grid, will most surely be more expensive than an urban settlement with a high population. Therefore, other distribution technology solutions other than the national grid, may prove to be more affordable in certain areas of the country. As shown in the result obtained by ONSSET, the cost of electricity depends highly on the distribution technology available in the area. The areas supplied by standalone systems, such as Solar PV, shows a higher electricity cost than those supplied by the grid. Naturally, this is a consequence of the high costs of grid expansion in these areas.

If the energy services can be considered to be affordable, there reliability is another issue. As mentioned earlier, CSP is a large contributor to the generation mix, which makes the electricity distributed by the grid sensitive to weather conditions. This weather sensitivity is amplified by the contributions of wind power. Furthermore, there is an uncertainty concerning the contributions of coal and natural gas. If these fuels would get more expensive in the future due to decreasing reserves or introduced costs related to GHG emissions; it could affect the cost and reliability of the electricity production. Furthermore, the reliability of the off-grid solutions also holds some uncertainties based on the same reasons as mentioned above, and additionally the systems may be more sensitive to power cuts since they will require individual backup systems.

6.3 Expansion of hydropower as a solution

Because of the abundant water resource in Ghana, it is possible to install large scale hydropower plants. There are also several constructions of new small hydropower plants. Meanwhile, this kind of power plants seems to be limited, as much of the potential is already installed. Also, the electricity demand will be rising from 45 PJ in 2016 to 500 PJ in 2050, which is an increase by ten times. In the results for year after 2020, there is a clear trend of a decreasing share of hydropower. In conjunction with increased population and higher electricity need, hydropower can no longer satisfied the electricity demand in the country.

Hydropower technology can be viewed as a mature technology in Ghana and was established very early. The fact that the country has continued to choose this technology can be interpreted as a sign that it is an affordable option. Meanwhile, the hydroelectric power and thereby the grid price is deeply affected by different climates. As assuming, there should be humid, dry and business as usual scenarios, the grid price varies between 0.06 and 0.08 USD/kWh.

To sum it up, hydropower is an important part of the solution for the electricity system in Ghana to develop, especially in the medium-term, while the climate is stable and a somewhat normal rainfall pattern can be expected. But with a dryer climate and above all a considerably increasing demand, the share of the production supported by hydropower will be marginal.

6.3.1 Vulnerability of hydropower

Although hydropower is dominant in the present electricity generation, the share of it will decrease significantly in the future, as the total generation is increased. This is mostly a consequence of a rising electricity demand that can no longer be satisfied with generation from hydropower due to a insufficient potential in the country to increase the production.

The result also shows a greater share of hydropower in the generation mix in the humid scenario than in the dry scenario. Clearly this is an effect of the technology being more economically profitable in a humid climate than in a climate where the precipitation decreases and as a result the capacity factor of the power plants. The result does not take into consideration the potential conflict between water need for power production and water need for agriculture. In a dryer climate, the need for water for irrigation could potentially increase which would strengthen the picture of hydroelectric power being a less favorable technology for Ghana in the future.

In the model, the electricity generation mix was projected for both a dryer climate and a more humid climate. This was done even though the more accepted scenario today is that Ghana is projected to get drier in the future. If this projection is correct, it could result in that investments in an expanding hydroelectric power sector could prove to be unprofitable already in the medium-term.

6.3.2 CO₂ Emissions under different scenarios due to the share of hydropower

As the result shows, there is a clear difference in emissions between the three scenarios, especially in the short term. The pattern changes in the end of the modelling period, where the differences between the scenarios decrease. Especially between BAU and dry, where the difference is negligible during the last years. This change could come as a result of the decreasing share of hydropower, since the climate change is assumed to affect only this technology in the model.

In the early years, it is expected that hydropower will have significant share of the generation regardless the scenario, thus its impact on the total emission ratio should be more noticeable. For the humid scenario, hydropower has a production very near to its maximum potential throughout the whole timeframe, but in conjunction with the speedily increased total generation, its impact on the emissions has less effect as its share decreases. Therefore, the humid scenario has the same trend when it comes to emissions as the other two scenarios with less hydropower production.

In short, the OSeMOSYS model has chosen to invest in other technologies such as coal power plants when the possibilities of generating hydroelectric power, due to climate changes in the dry scenario, are unavailable. Thus, the emissions resulting from the electricity production, are relatively high during the whole period, compared with the humid scenario. In the end of modelling period, the share of hydropower decreases and its effect on the emission appears to have less effect, and therefore all three scenarios show the same emission trend.

6.4 Source criticism

Fuel prices are often important factors affecting the total cost of using a technology. However, it is difficult to project the price, as many factors such as politics and the economic situation could affect it. Thus, the fuel prices applied in this case study are assumed to be constant, which is the latest actual delivered price. This value was chosen to ensure that the accuracy of the fuel to be as correct as possible. Though it remains that if this assumption is poor, it will constitute to potential uncertainty in the results.

Another question worth discussing is the date on which of new hydropower plants will come online. There is not enough information about all the new small hydropower plants, including which year they will be able to put into use and generate electricity. Therefore, the construction time is assumed to be ten years, that is to say the year at which the uncertain power plants will be completed in 2027. This assumption has obvious effect on the model's choice of technology, and may lead to a misleading projection of the electricity generation after this year.

The Energy Commission of Ghana is a reliable authority and providing information regarding energy, therefore it is often used in this project to obtain information within energy sector, such as energy production and usage. However, it is a clear uncertainty to rely too heavily on one source, and in many situations this study is lacking other sources to confirm the information.

A necessary step to develop different climate scenarios in the models, was to project the capacity factors for the hydropower technologies, since these are the parameters that are expected to be affected by a changing climate. These values cannot be assumed to be completely accurate since they are based on a variety of assumptions. The long time projected, also makes it harder to make correct assumptions on something that is based on such a complex aspect as climate change.

6.5 Future research

The research done in this study has been done to give a broad overview of the electricity sector in Ghana. This study focuses on the links between climate, water and energy, with a main focus on climate change and its impact on electricity generation from hydropower. To deepen the understanding of the interlinkages between these three sectors, the authors have identified a few areas to be of great interest and importance for further study to make it possible to generate better results.

- Include more possible electricity generation technologies in the modeling, such as waste to energy, nuclear power, wave energy and other similar technologies.
- Include the energy use for the food sector in the analyses and the possibilities to develop and electrify this sector and to reduce its GHG emissions.
- Make a deeper analysis of the effects of climate change on the whole electricity system to see if other technologies, such as wind and solar, will in any way be affected by the changes.
- Research the interlinkages between hydropower and agriculture and the impacts of a climate change.
- Include possibility of electricity generation in more undeveloped rural parts of Ghana, such as the north of Ghana.

7. Conclusion

What can be concluded from the discussion above, and from the report overall, is that there is a credible possibility of Ghana having to transform its electricity sector in a rather drastic way to be able to meet the steadily increasing demand for electricity until 2050. What may also be concluded, is that in order to transform the electricity sector to meet future demand a large share of the electricity has to be generated from renewable energy sources such as solar and wind energy. To fulfill the goal of affordable and reliable electricity, it may also be concluded that the generation mix has to consist of other technologies as well, due to the RE technologies' reliance on weather or climate factors. From the result, it is also clear that the electricity distribution will have to be developed in order to meet the increased demand. This will have to be done as a combination of expanding and updating the grid as well as implementing other distribution technology solutions such as stand-alone systems and mini-grids.

Furthermore, it can be concluded from the discussion that the amount of electricity generated from CSP technology contribution would benefit from further investigation, due to the great uncertainties concerning the technology and its costs and performance. Despite these uncertainties concerning CSP, it is evident that solar energy can play an important role in the future development of the electricity generation system of Ghana. –. After analyzing the results, it can also be concluded that an expansion of the hydropower sector will not be a large part of the solution for the development of the electricity sector in Ghana. This is partly because of the limited potential of hydropower expansion in the country due to the already well developed and expanded hydropower sector in the country. It is also a result of the technology's vulnerability to climate change and especially great sensitivity to changing precipitation patterns.

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